BUILDINGS & CITIES

Increasing civic resilience in urban living labs: city authorities' roles

SPECIAL COLLECTION: LIVING LABS: AGENTS FOR CHANGE

RESEARCH

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ubiquity press

ABSTRACT

When pursuing change through urban living labs (ULLs), local authorities encounter a dual imperative: they must act as drivers of change while remaining open to adaptation themselves. Navigating this tension requires a range of context-connecting practices among stakeholders. These dynamic interactions between citizens and city authorities are investigated in the transformation of an industrial area into a sustainable urban district, thereby foregrounding the civic dimension within the broader resilience discourse. The case study focuses on a ULL in the city district of Hiedanranta, Tampere, Finland, where citizen involvement and fostering bottom-up initiatives has been central. The analysis draws on ethnographic, interview and documentary materials from action research conducted between 2016 and 2025. This long-term data allows a question to be asked: how can city authorities increase civic resilience through different phases of a ULL? To understand the temporalities of ULLs, a system life-cycle view is applied, including the early phase of setting the stage, which can efficiently integrate citizens into the stakeholder network. Persistent collaboration with citizens was crucial in setting the tone for the entire area development process. These efforts were instrumental in realising new design concepts, such as Nordic superblocks, and highlighting the importance of shared spaces.

POLICY RELEVANCE

City authorities and citizens can and need to support each other to foster change within and through ULLs. Attention should be given to practices that connect citizens and city authorities for collaboration and mutual learning, thereby building the agency of both. These practices involve bringing actors and contexts together, creating persistent tactics and catalysing positive impact loops. Specifically, they include reinviting stakeholders, developing shared spaces, establishing new legal entities and recognising emerging ideas and actors. It is crucial for multi-actor groups, including citizens, to start working together very early on to invent and define ULL projects. Throughout the phases of a ULL, there are various ways for city authorities to enhance citizen involvement and agency. When these practices fail, there is a high risk of losing crucial actors, their knowledge, and the processes of change they contribute to.

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KEYWORDS:
urban living labs; resilience; urban governance; placemaking; co-creation; urban planning; community participation; civic engagement; Nordic superblocks; Finland

TO CITE THIS ARTICLE:
Alatalo, E., Laine, M., & Kyrönviita, M. (2025). Increasing civic resilience in urban living labs: city authorities' roles. *Buildings and Cities*, 6(1), pp. 917–933. DOI: https://doi.org/10.5334/bc.610

DOI: 10.5334/bc.610

1. INTRODUCTION

In a time of climate crisis, severe biodiversity loss and complex social issues, cities face challenges that require innovative governance solutions. In this context, local governments have become supportive of urban experimentation, viewing the city as a platform for open, collaborative innovation processes aimed at promoting systemic change towards sustainability. A common form of such governance is urban living labs (ULLs). These labs can be broadly defined as places where local governments, citizens, businesses, universities and other actors such as social organisations form partnerships to develop and test new urban solutions in real-life contexts (Juujärvi & Pesso 2013; Kronsell & Mukhtar-Landgren 2018; Turku *et al.* 2022).

A distinct feature of ULLs as a form of experimental governance is their explicit learning function (Bulkeley *et al.* 2019). ULLs aim to engage actors across various fields and sectors in co-creating and disseminating knowledge to facilitate broader policy learning processes (Baxter 2022; Mukhtar-Landgren *et al.* 2019). This reflects the growing role of local governments as enablers of partnership-based models where they, according to Bulkeley and Betsill (2013: 141), 'participate but do not have an explicit leading role' (see Mukhtar-Landgren *et al.* 2019: 721). This paper examines one such co-creative ULL, where the city organisation collaborates with multiple actors to develop innovative solutions for transforming a former industrial area into a sustainable city district. The analysis provides insights into the various ways city authorities in a ULL can participate in learning and change processes.

Recent literature on ULLs highlights their ability to foster broad societal participation and to develop hybrid understandings of complex urban issues and among differing interests (Rizzo *et al.* 2020; Robazza 2024). ULLs are also seen to enhance society's capacity to adapt to challenges by empowering communities and supporting the development of participants' skills and knowledge across different societal sectors (Robazza 2024). This suggests that ULLs can potentially increase resilience – understood as the adaptive and learning capacity of individuals, communities and institutions – to not only maintain the urban system but also reinvent it in the face of current challenges (Othengrafen *et al.* 2024: 24–25).

There is a shift going on from the resilience discourse of responding to varied societal emergencies towards the approaches that understand resilience to include the aspects of empowering people (MacKinnon & Derickson 2013; Petrescu *et al.* 2016). In this shift, the concept of civic resilience comes to the fore, where the learning capacity that compounds the resilience (Othengrafen *et al.* 2024: 24–25) is especially about citizens learning to act, *i.e.* learning agency in the change processes (Antaki & Petrescu 2023). This is not an easy task for citizens (Alatalo *et al.* 2023b; Portelli & McMahon 2004), which in the context of the ULLs, where the city organisation plays the enabler's role, puts pressure on the city authorities to facilitate the agency learning processes of people and, thus, to facilitate the increase of civic resilience. However, as framed in the following chapter, city authorities also crucially need citizens from early on to be active stakeholders in the ULL change processes.

This leads to the research question under investigation: How can city authorities increase civic resilience through different phases of a ULL?

When exploring this question, civic resilience is demonstrated especially through the focus on how citizens learn to act and be part of the change processes. Although the empirical material of this paper does not delve into the civic learning processes as such, it shows how the agency of citizens may grow over time, supported by the city authorities. This sheds light on the roles of city governments as long-term hosts and guides of transition.

2. EMBEDDING CITIZENS INTO LONG-TERM CHANGE PROCESSES

2.1 CONTEXT-CONNECTING PRACTICES FOR DRIVING AND ADAPTING TO CHANGE

As part of studying a project in the Hiedanranta ULL in Tampere, Finland, a promising context-connecting model for projecting sustainability transitions has been developed (Lehtimäki *et al.*

2023). The model's idea is based on the observation that cities face a significant dual-context challenge when aiming for sustainability transitions: they need to both change and drive change. Cities play a central role in leading sustainability transitions (Frantzeskaki *et al.* 2017). They are expected to act as reliable long-term hosts and guides of transition, despite being organisationally siloed and fragmented, with diverse interests and opposing aims (Frantzeskaki *et al.* 2017). Additionally, the city as an organisation must evolve, adopting the best learnings as the ULL experiments progress. Even short-term projects can equip their host organisations to facilitate the transition (Lehtimäki *et al.* 2023). The key lies in context-connecting practices that focus on both the host organisation and the external experimental environment where the sustainability transition takes place. Thus, in the concept of context-connecting practices, the word context refers to the two contexts of change: to the change of the enabler organisation and to the change it drives, with the focus on the need for these two contexts to be connected. In this approach, the attention is on stakeholder collaboration that builds trust and resilience step by step (Lehtimäki *et al.* 2023).

The context-connecting model offers a nuanced opportunity to study the formation of civic resilience between city authorities of a ULL and its external actors, such as residents. Lehtimäki *et al.* (2023: 7–10) define five context-connecting practices:

1. bringing actors and contexts together
2. creating persistent tactics
3. reflecting
4. re-evaluating and choosing the course of action
5. catalysing positive impact loops.

This research focuses particularly on practices 1, 2 and 5, explained in detail below, to better understand the practices where city authorities may increase civic resilience through supporting citizens in learning to act and be part of change processes.

The practice of bringing actors and contexts together (1) serves a dual purpose: creating a shared vision and developing the new abilities needed to realise it. This requires dedicating time to build momentum for the common cause and to learn together. It also involves encouraging individuals to take on new roles and interacting across previous departmental, professional, hierarchical or societal silos. The goal of creating persistent tactics (2) is to foster commitment and establish long-lasting, transition driving practices among stakeholders. This involves repeatedly emphasising shared goals and having the courage to challenge conventional practices that hinder change. Catalysing positive impact loops (5) focuses on building a long-term support system for the stakeholder group. To achieve this, it is crucial to recognise weak signals and help them gain momentum, leading to wave-like progress (Lehtimäki *et al.* 2023).

By focusing on the context-connecting practices between citizens and city authorities, the practice theories of intermediation (Hernberg & Hyysalo 2024; Kivimaa *et al.* 2019) related to ULLs are explored. The development of the roles of different intermediaries has been shown to be typical in change processes, varying in different phases, but more research is needed (Hernberg & Hyysalo 2024; Kivimaa *et al.* 2019). To grasp the development of the roles of the intermediaries, here the city authorities, a framework to understand the different phases and temporalities of a ULL is needed. In the following, one such framework that can be discussed well with both the ideas of context-connecting practices (Lehtimäki *et al.* 2023) and the case study is presented.

2.2 SYSTEM LIFE-CYCLE VIEW TO UNDERSTAND THE TEMPORALITIES OF ULLS

This research addresses the need to better understand the temporal dynamics of ULLs (Overdiek 2024; Turku *et al.* 2022). There are temporal models of living labs (LLs), such as the ISLE model, which looks into how a transition from current system to a preferred one may be done (De Lille & Overdiek 2021; Overdiek 2024). This is a typical example of quite abstract temporal LL models that,

DOI: 10.5334/bc.610

despite accentuating the connection to place and local experiments, lack the temporal connection to the processual legal and political realities of developing and building a city. Urban development is a phased process, and district-sized ULLs follow these phases. Thus, this research considers the phases of a ULL from the perspective of the phases of building an urban district. The opportunities to promote civic resilience vary in each of these phases.

In studying complex building projects, a system life-cycle perspective has been adopted to understand how value is created among the multiple stakeholders and organisations during the different phases of building (Artto *et al.* 2016). This article applies this perspective by focusing on citizens as stakeholders and civic resilience as the value created.

The system life-cycle perspective suggests that a building project generates outcomes that continue to operate and add value even decades after the project phase has ended (Artto *et al.* 2016). The key to creating value in project management lies in how the variant stakeholders' work is integrated within the multi-organisational system (Morris 2013). This integration has been examined by aligning the project phase, which includes the design and building, with the operation phase, which means living in the new urban area and looking into the activities that bridge these two phases. In doing so, the importance of establishing a functional network of stakeholders as early on as possible has been recognised, and this has been proved to significantly enhance value throughout the entire system life cycle (Artto *et al.* 2016). It has also been noted that there is a high risk of losing crucial knowledge, actors and their practices when transitioning from one phase to another (Artto *et al.* 2016). The context-connecting practice of creating persistent tactics helps to bridge these gaps (Lehtimäki *et al.* 2023).

To understand the temporal dynamics of ULLs, this paper first examines the 'setting of the stage' phase, which precedes the project and operation phases (Artto *et al.* 2016). The beginning of the project phase is explored by focusing on the area's design and the implementation of the block bidding process. Therefore, the phases investigated in this article are:

- setting the stage
- collaborative design
- implementation.

The collaborative design phase is emphasised in the analysis because of the richness of the research material. By examining these phases, the paper demonstrates how civic resilience is shaped throughout the project system life cycle from the very beginning.

3. DATA COLLECTION AND ANALYSIS

This paper is based on an intensive nine-year period of ethnographic field work (Okely 2012), participatory observation (Clark *et al.* 2009) and participatory action research (Chevalier 2019) from 2016 to 2025. The research data comprise 18 interviews with city authorities carried out between 2016 and 2024. In this context, city authorities are understood to include both municipal civil servants and external experts working for the city in Hiedanranta development. The individual interviews with municipal officials have been categorised as follows: project managers (4), property managers (2) and planners (2). The interviewed external experts, in turn, fall into the following categories: project workers (2), communications specialists (1), consultants (1) and architects (2). Furthermore, four group interviews were carried out: two involving two planners each, one with two architects, and one with two directors from the city-owned company Hiedanrannan Kehitys Oy, founded in 2019. The data also include 31 recordings of interdisciplinary groups working on the development of Hiedanranta in workshops organised for that purpose, 25 of which included citizen participants. Two researcher field diaries from a citizen participation experiment related to the construction of a public sauna are also utilised. Supplementary data, including public planning documents, reports and website screenshots, are used to further illustrate the case for the reader.

To analyse the data, a collaborative thematic analysis was conducted (Braun & Clarke 2006; Hemphill 2018), with a particular emphasis on identifying patterns that reflect the emergence

of citizen agency and the supportive roles of city authorities in urban change processes. Drawing on the notion of citizens learning to act – *i.e.* developing agency within transformative contexts (Antaki & Petrescu 2023) – the analysis first identified key moments of agency formation, which were then examined in greater depth through collaboratively developed coding schemes (Hemphill 2018).

4. THE CASE OF HIEDANRANTA

The Hiedanranta ULL in Tampere, Finland, is situated in a former pulp industry area that is currently being transformed into a sustainable city district along a new tramline, offering in the future housing for 18,500 people and jobs for 6,000 (The City of Tampere 2025). This whole district-size ULL is led by the city organisation, with the goal of developing innovative solutions and sustainable infrastructures. A key aspect of its co-creative approach is the involvement of citizens and the promotion of bottom-up initiatives that contribute to both the operation of the ULL and the development of the city district. The lab experiments with novel practical solutions as well as alternative modes of leadership and ownership (see Bulkeley *et al.* 2019).

Tampere is renowned for its industrial history. The city began to grow rapidly in the mid-19th century as textile manufacturing, engineering, and wood processing industries developed along the banks of Tammerkoski rapid, which provided a favourable energy supply. A significant part of the city's industrial heritage is also found in Hiedanranta, which has a hundred-year history of wood processing. However, in 2008, the M-real company shut down the production, leaving the factory buildings mostly empty until 2014, when the city of Tampere purchased the area.

Recently, Tampere has experienced even more rapid growth than during its early industrialisation. With over 5,000 new residents in 2024, the city's population has surpassed 260,000 (Statistics Finland 2025). Such a growth creates pressure for sustainable urban development and justifies major infrastructure projects, such as building a new tramway system and acquiring land for infill development. These initiatives are intertwined in Hiedanranta, where the goal is to transform a former privately owned brownfield site into a sustainable, dense and attractive urban hub of western Tampere (Hiedanranta Master Plan 2020).

The sustainability goals of Hiedanranta are ambitious, encompassing CO_2 negativity, a circular economy, innovative energy systems, digital solutions, and sustainable transportation supporting walkability, cycling and the use of public transport. Additionally, art and community programmes are implemented to support events, cultural facilities and communities, aiming to create a lively, experiential area with an international appeal (Hiedanranta Master Plan 2020). This approach reflects the city's commitment to involving citizens in the area's development from the outset. Rather than keeping the gates closed, the city began experimenting by opening the former industrial area to citizens and various actors. This inclusive strategy has engaged researchers, urban activists and citizens who became actively involved in the area's development and planning.

5. INCREASING CIVIC RESILIENCE IN THE THREE PHASES OF ULL

As stated earlier, in city-driven ULLs involving city authorities, such as planners and local project coordinators, there is a need to balance in their double role of driving the change in the urban context and to change the practices of the city organisation itself (Lehtimäki *et al.* 2023). Another challenge for city authorities is ensuring that both technical and social innovation occur. For this, it has been proposed that it would be useful to accentuate one or the other in different phases of a ULL, and that intentions towards social innovations should precede the technical ones (Overdiek 2024). Based on these balancing challenges, the roles of city authorities in the Hiedanranta case are clarified in Table 1. The three ULL phases are discussed in detail below.

	PHASE 1: SETTING THE STAGE (2015–2019)	PHASE 2: COLLABORATIVE DESIGN (2017–2022)	PHASE 3: IMPLEMENTATION (2022–ONGOING)
Main activities	Taking vacant spaces back into use with citizens and small enterprises	Creating the spatial and functional concept of the area, until first city plans	Guiding and forcing collaborations to make innovations real
Main roles of city authorities balancing between social and technical innovations	Supporting social innovation: new collaborations, experiments	Supporting technical innovation: including new spatial, functional and technical innovations	Supporting technical innovation that will support social innovation in the future
Main roles of city authorities balancing between driving change and changing themselves	Changing themselves as becoming open and courageous partners in several experiments, of which some also failed, creating new legal entities to do so	Driving the change by being the leaders of the collaborative design and consolidating the results into legal documents	Changing themselves by creating new legal entities to guide and manage the building of the area until future residents and other users arrive

Table 1: The phases of Hiedanranta ULL and the main roles of the city authorities.

5.1 SETTING THE STAGE FOR COLLABORATIVE PLANNING

The City of Tampere adopted a uniquely participatory and experimental approach to developing Hiedanranta. The aim was to create the city district collaboratively with residents, businesses and communities. The area was opened in 2015, and vacant factory spaces were rented to businesses, social organisations, artists and cultural actors. This open-minded development was organised under the Hiedanranta Development Programme, a 'project-like' structure with dedicated project management. The development programme comprised three complementary subsectors: Innovative Hiedanranta, Temporary Hiedanranta and Planning.

Innovative Hiedanranta served as a development platform for experiments and projects promoting smart technology, sustainability, and circular economy solutions emphasising cooperation with city residents, businesses, research institutions and other organisations. As one of the project managers noted, the genuine goal was 'to find, with the help of participatory and innovative platform-like processes, new ways and models for planning and implementing such a sustainable and intelligent city district' (Project manager, 22 February 2019). This approach successfully harnessed valuable expertise and knowledge, with companies, universities and research organisations such as VTT Technical Research Centre of Finland and Finnish Environment Institute engaging in Hiedanranta development. Examples included prototyped vertical gardening, biochar and low-tech solar energy storage. Additionally, some firms that are now internationally recognised, such as Polar Night Energy and Carbofex, were established during this period.

Temporary Hiedanranta brought actors and contexts together by versatile cultural activities led by citizens, businesses and communities to develop a vibrant and sustainable urban culture. The actors had rather free hands to pursue their activities and use the vacant factory spaces that the Hiedanranta Development Programme rented out. A decisive factor in engaging different actors to Hiedanranta was progressive rent, which was very low in the beginning. The process of taking vacant spaces back into use was done in a wise way: the actors had time to grow and develop their resources as well sources of income while also modifying spaces to fit their needs (see also Alatalo *et al.* 2019; Kyrönviita & Leino 2025). All that created rich processes of increasing citizens' agency and thus civic resilience in relation to specific spaces.

As part of the rental agreements, the actors were required to organise events and promote Hiedanranta to increase its attractiveness in this early phase. However, generating the impression of experimentation and coolness through temporary uses was not the only purpose of Temporary Hiedanranta (Fabian & Samson 2015; Madanipour 2018). It was understood as 'a really good way to start engaging people and nurture forms of long-term involvement' (Project manager, 7 April 2021). That worked as small businesses, artists and artisans as well as grassroots cultural and social organisations took root in the area. The early engagement of citizens into local change

created relationships, of which some lasted through the ULL phases ahead, increasing civic resilience and contributing also to the socially sustainable development of Hiedanranta (see also Rikala *et al.* 2023).

Along with the experiments and cultural activities, the planning process was started in an exceptionally dialogical manner. A wide range of stakeholders were engaged already long before the legal requirements of the planning process. In 2015, the city planners and university researchers collaboratively conducted an ideation process leading to the Development Vision of Hiedanranta, which was open and free for everyone to use (Lehtovuori *et al.* 2016). Subsequently, the Hiedanranta Development Programme, with the help of consulting firms Kuowi and Uusi Kaupunki Kollektiivi Oy facilitated series of round table discussions and events with companies and citizens (Korpinen & Pulkkinen 2015; Virkkala 2016). Both consulting firms promoted open collaboration and they, together with the communications specialists as well as examples from Helsinki and Jyväskylä, convinced the hesitating project managers of openness and 'that it is the way of the past world if we operate in a close area and do planning and preparation' (Project manager, 12 August 2019).

Also, unconventional approaches such as Hiedanranta garden parties were organised to gather ideas and wishes from residents to support planning and development. The first garden parties, held in 2016, attracted thousands of enthusiastic people. They were not just visiting the area as mere spectators but actively producing their own environment by organising activities ranging from concerts and children's workshops to cafes and guided architecture tours (see Crossan *et al.* 2016). Such an approach, which allows people to develop and engage community resources in an area undergoing massive urban transformation, can be seen to increase civic resilience (Magis 2010; Othengrafen *et al.* 2024). People's engagement had a profound effect on the city officials. They were accustomed to people participating in planning only if there was something to complain about, but the garden parties proved this wrong. People shared their ideas and initiatives and participated with enthusiasm in organising the garden parties and other activities that were seen to contribute to the planning and development. As one of the planners described, 'it [citizen participation] has been one of the goals that we all have promoted all the time' (Planner, 13 March 2020).

> I still think that the new technologies, culture, and urban planning are the core of Hiedanranta. It is not just urban planning, it is not just culture, and it is not just new projects but together they form an interesting whole. That I think is the real value.
>
> (Project manager, 12 August 2019)

The way the Hiedanranta Development Programme was organised – as a project with a flat organisation – enabled novel practices to emerge in the context of urban planning. The development programme created linkages between micro-activities in Hiedanranta and macro-context of the project in the city organisation and in the context of urban planning. When one of the project managers described the importance of the Hiedanranta activities in the broader context, he wanted to point out how those activities 'began to drive the whole thinking of how general planning was conducted and thought. It was the idea that such [connecting] activities could stay there' (Project manager, 7 April 2021). It can be said that the Hiedanranta Development Programme worked as an intermediary allowing experimentation, a bottom-up approach and flexibility while also fostering long-term strategic changes, which became visible in later phases as new legal entities to further enable change in an otherwise siloed and departmentalised city organisation (see also Lehtimäki *et al.* 2023).

Encounters of diverse people were supported in the early phase by the selection of actors. For example, Sopimusvuori is an association that offers social rehabilitation and rehabilitating work services, with a special focus on mental health. They had two main activities in Hiedanranta that offered essential services also for other early arrivals. First, they formed a group of people, mostly older men, who were skilled with building and maintenance. The group was easily contactable whenever relevant local problems needed fixing. They were like district janitors. Second, Sopimusvuori provided a coffeehouse. Relying on sales, it would have been almost impossible to

keep a coffeehouse profitable in an area that is just about to start to draw the attention of the people. Nevertheless, as part of the budget came from the rehabilitation of the people, the profit needed from selling coffee and food was minimal. The existence of the coffeehouse since the opening of the area was however crucial, as it created an important social gathering place. Most importantly, it created new possibilities for building trust, when strong pioneering groups met with people in vulnerable life situations, in a way where these vulnerable people were not the objects of help but could provide help and support for others (see also Rikala *et al.* 2023). The example of Sopimusvuori illustrates the catalytic possibility that emerges from changing the usual roles of interaction in the context-connecting practices (Lehtimäki *et al.* 2023).

There were also moments when the city failed to support civic resilience, instead hindering people's agency and pushing them out. Some processes of taking vacant spaces back into use got blocked, because the cost estimates for the renovations were too high because they were given on the basis of standards that were far higher than the pioneering groups (*e.g.* a makerspace led by the university) needed and were capable of paying rent for. When the building that had housed, for example, circus and large electronic music events was demolished to make way for the tramway, people, their practices and their knowledge were lost, and some communities did not survive the change. There have also been start-up companies (*e.g.* Carbofex) that were not accommodated in the area while it formalised, even though it could have been possible. Some pioneering groups also became exhausted by the demand of organising events and presenting their activities to the public.

From the perspective of citizens' agency, this phase of Hiedanranta ULL illustrates how participatory urban experimentation can empower residents to shape their environment and engage in longer term transformation processes. Through open access to spaces, low-threshold involvement, and dialogical planning practices – including cultural activities, experimental projects and ideation events – citizens were able to build capacity, form communities and contribute to urban planning. These practices fostered civic resilience by enabling citizens not only to participate but to lead and co-create, although moments of institutional rigidity also revealed the fragility of such agency when not adequately supported. Therefore, a more collaborative approach was adopted.

5.2 COLLABORATIVE APPROACH TO DESIGN

Some of the outcomes from the phase of setting the stage were concretised into documents (*e.g.* Korpinen & Pulkkinen 2015; Lehtovuori *et al.* 2016; Virkkala 2016) that served as background material to international ideas competition. The goal of the competition, held in 2016, was to find the overall spatial and functional concept of the area. In relation to the system life-cycle view on complex building projects (Artto *et al.* 2016), it is here that the project phase commences, as the project concept starts taking more precise forms through the idea competition.

In the competition call one could find themes related to civic resilience seen as inclusion, such as aims for wide stakeholder participation during the design phase or aims for urban structure that would encourage meetings of future dwellers (The City of Tampere 2016). The idea competition was won by two architectural collectives (The City of Tampere 2017a). At the time, the experimental and open attitude of the Hiedanranta Development Programme had drawn the attention of a research project that focused on citizens' role in urban transformations. An opportunity to act for and do research about citizen participation was seized. The Hiedanranta Development Programme was eager to collaborate, and a series of five co-design workshops was organised. It became like a joint festival, before any legal planning process had begun (Alatalo *et al.* 2017).

Starting from ideas of organising a citizens' panel that would critique the newly published winning entries, this led to facilitated workshops, where citizens, planners, traffic designers, architects and researchers worked all together over the proposed ideas to develop them further. The co-design workshops allowed long and deep discussions: each lasted for three hours and ended with a shared meal. Three of the workshops were organised in sequential evenings. Altogether there were 202 participants in the first four workshops, of whom 121 were citizens. The fifth workshop was focused for researchers. One essential element was that these workshops were the first moments

DOI: 10.5334/bc.610

when all experts and future designers of the area started to work together. As part of the starting discussions, stakeholder groups and citizens were involved with inputting their experiential, personal and situated knowledge. Thus, these workshops did set the mood and expectations for the whole legal planning process to come (see also Alatalo *et al.* 2017; Sjöblom *et al.* 2021).

From these workshops emerged themes that gained broader support and ignited discussion in relation to the future of Hiedanranta. One of the strongest concepts was named the superblock, referring to a new scale of collaboration on a block level or between several blocks. This was seen to create better quality and more versatile shared spaces for residents, and to provide tools for lively streetscape and energy communities (Alatalo *et al.* 2017). In current Finnish urban planning, the design of neighbourhoods is done at the master plan level. Then the neighbourhoods are realised in a developer-driven way in stages by constructing stand-alone buildings one by one on separate lots. Thus, planning shared spaces at the block level has not really been possible before. Citizens seized the opportunity with enthusiasm and brought visible their wishes and needs for shared spaces that also support inclusion and thus civic resilience.

> It makes no sense to build apartment buildings with studios and saunas. Let's make a communal sauna and, of course, there can be something that can be reserved for private use [...] Some communal spaces may be good to have in your own housing company. Some can be spread more widely in the block if you think about it.
>
> (A citizen in the Bustling Blocks –workshop, 27 April 2017)

There were already tentative elements of the superblock concept (later named as Nordic superblocks) in the competition call, as well as in the drafts in the entries to the competition (The City of Tampere 2016; 2017a). However, it was the workshops that enabled the concept to evolve and gain sufficient support, giving the architects the confidence to further develop the idea – ultimately leading to political acceptance.

> I would see as the greatest value in the workshops that they did set a certain state of will [...] For me those workshops were one, quite important part, especially as in them we genuinely delved into the issues, and there were relatively many participants. So, they [the workshops] kind of had credibility and in a way, it was shown that hey, this is the right thing, and this is what we want.
>
> (An architect, 23 April 2018)

The workshops formed a citizen–expert coalition that was needed to create new alternatives to the professional modes of practice usually dominated by market, creatives and regulations (Carmona 2009; Sjöblom *et al.* 2021: 118). Various workshops and the participation of different groups have played a big role in the planning of Hiedanranta. Through these the city planners feel that they are constantly absorbing and filtering new information about how Hiedanranta should be developed. Their concern is that, in a long and multifaceted process, something always falls by the wayside and gets less attention. That's why they feel it is important that the same themes are brought up again and again to public discussion and the citizens can check the direction of the design and share their views along the way. In other words, planners have noted the need for context-connecting practices (Lehtimäki *et al.* 2023) with citizens throughout the project of realising a city district.

> They [the citizens] have to be made to come again and again [to participate into the planning], so that they sort of check whether things have been kept up with all the time. Because it's quite possible that we forget something, because we can't digest everything just like that.
>
> (A planner, 11 September 2017)

Various communities that had settled in the area and were evolving at the time were together recognised as assets (Alatalo *et al.* 2017). For example, over the course of a decade, the skateboarders evolved from a DIY collective into official partners of the city collaborating in

education, urban development and the marketing of the city (Kyrönviita & Leino 2025; Kyrönviita & Wallin 2022).

The Hiedanranta Structure Plan (2017b) was the next legal document in which the future of the area was presented, including also a page mentioning the workshop series with a promise that their results are included in the plans of the area. However, the results of the workshops scarcely appear in the document, as it deals with design principles on such a large scale (see also Sjöblom *et al.* 2021: 125). While continuing towards the Hiedanranta Master Plan (2020) that would guide the local detailed planning, there was one large citizen workshop facilitated by the same researchers in spring 2018. This workshop nevertheless had more of an atmosphere of informing citizens with plans evolving on a tight timetable. Even if there was further research ordered by the development programme on the possible application of the Nordic superblocks to Hiedanranta, it started to seem that the idea was getting lost in the planning process of the area.

Although Hiedanranta had no residents, at this point the different groups and citizen communities that had started to take vacant premises back to use had become quite stable. Most of them had had enough time to develop their own communities, repair their places in buildings and figure out their income models (Alatalo *et al.* 2023a). These were the strong and talented citizen groups who knew how to seize the opportunities in the vacant spaces. However, they did not have interactions with each other. In addition, there was still room for new citizen groups in the vacant buildings, if they only knew how to join the process. This moment exemplifies how different context-connecting practices (Lehtimäki *et al.* 2023) are needed at different phases of the ULL. Time was ripe for another community building experiment. Facilitated by the same researchers and supported by the Hiedanranta Development Programme with building materials and free rent of land, an open process of designing, building and maintaining a public sauna in Hiedanranta was started. Anyone was able to join the process and also use the sauna, once ready, anytime, free of charge (Alatalo *et al.* 2023a).

It became a struggle and a learning process for all stakeholders over time. Fortunately, the efforts resulted into a place that spatially joined different pieces of the area, got separate local groups to achieve something concrete together and brought new kinds of proactive citizen participants to the area development (Alatalo *et al.* 2023a). In Finnish tradition, sauna is a place where people meet on equal terms. With your clothes, you leave all the marks of your societal status out from the sauna. People of different ages and backgrounds sit next to each other and personal discussions easily emerge. In the sauna of Hiedanranta, people released from prison enjoy steam together with professors, and engineers with circus artists alike. The success of sauna talks about the importance of context-connecting practices (Lehtimäki *et al.* 2023) that create social spaces where diverse people can interact.

In this phase, citizens played a formative role in shaping the Hiedanranta development through early co-design workshops, where their experiential knowledge was treated as equal to that of planners and experts, setting the tone for inclusive planning. These workshops enabled citizens to articulate and advance concepts like the Nordic superblock, which challenged conventional planning norms and introduced new possibilities for shared spaces and civic resilience. Over time, citizens were able to seize opportunities for participation – such as building a public sauna – demonstrating their capacity to initiate, sustain and influence urban transformation through context-connecting practices. When approaching the realisation phase of the plan, citizens' agency faced a challenge such as the risk of ideas being diluted in lengthy planning processes and limited influence during formal implementation phase.

5.3 IMPLEMENTATION

The Hiedanranta development was reorganised in 2019, when the city-owned company Hiedanrannan Kehitys Oy (Hiedanranta Development Company) was established. The implementation phase followed as the first plot bidding was carried out autumn 2022. It covered four apartment blocks for 1,200 residents in an 8.2-hectare area from the northern section of the development area. Previous phases fed into the content of the bidding procurement. The second

DOI: 10.5334/bc.610

plot bidding followed a year after. The demands set by the city were ambitious, even 'foolhardy' (Director, Hiedanrannan Kehitys Oy, 13 February 2024) in relation to mainstream practice.

> Not everyone can do the same, but – we can adjust the height of the bar a little bit here and tighten the pace as we go forward. We can make the criteria more and more strict as overall performance [of the bidders] improves.
>
> (Director, Hiedanrannan Kehitys Oy, 13 February 2024)

The demands in the plot bidding competition had social and environmental aims. The social idea of the Nordic superblocks was carried out with a demand to construct good quality and varied communal spaces to the blocks. In Finland, communal spaces are usually constructed only for the use of the residents of one building. Thus, for example, realising a gym to one building and a sauna to another building that were shared between the residents of the block demanded novel collaboration from the designers and builders of the buildings. The implementation of ideas emerged with citizens in the previous phase of the ULL can be seen both supporting and valuing civic resilience, seen as inclusion and added agency of citizens. The operator of the communal spaces is going to be Hiedanrannan Palvelut Oy (Hiedanranta Service Company) that will be owned by the residents when the neighbourhood is finished. Hiedanrannan Palvelut Oy will provide a framework, 'a notebook for music' (Director, Hiedanrannan Kehitys Oy, 13 February 2024), according to which residents can then act together. Residents can decide what kind of activities are encouraged in communal spaces.

> The goal [...] in the case of shared spaces, should be that they actually provide [...] for the living environment quality and opportunities in a different way than the traditional model. And [...] another goal in my opinion should be to build less, i.e. concretely physically fewer spaces but still have at least the same level of living environment, or even, at best, a slightly better environment.
>
> (Director, Hiedanrannan Kehitys Oy, 13 February 2024)

> Hiedanrannan Palvelut Oy, in practice is built so that all the plots are committed to such a co-arrangement agreement, where it is agreed that everyone will contribute to the cost of making them [the shared spaces]. And then these jointly seen housing companies practically own the premises.
>
> (Director, Hiedanrannan Kehitys Oy, 13 February 2024)

Besides the communal spaces, Hiedanrannan Palvelut Oy will also operate a digital platform for residents. In the Hiedanranta plot bidding it was outlined to serve three purposes: as a curated interactive platform for different resident communities, as reservation calendar for communal spaces, but also for shared cars, and as a monitoring device for energy consumption and optimisation for apartment, building and neighbourhood level.

> Digital is also one of the tools by which it [shared spaces, services and functions] is made available to people [...] and of course there is the idea that it is also so for the information communication in the area and then for the formation of different communities in the area, to provide its one base.
>
> (Director, Hiedanrannan Kehitys Oy, 13 February 2024)

The implementation phase combines persistent tactics by which the core ideas of the Nordic superblocks are integrated to plot bidding demands as described above. It also contains a positive feedback loop in a form of binding co-arrangement agreement between forthcoming housing blocks. Unfortunately, the timings of the plot biddings were not ideal as the housing market was in decline in Finland owing to rising mortgage rates. Despite this the biddings were successful, as they had several entries. This was due to the existing identity of the area, formed by the events and other activities organised over the years, the industrial heritage, an attractive location by the lake, and the coming main tramline. In December 2023, as a result of the second bidding, four entries were selected to continue to the implementation. The construction work on them was

supposed to start autumn 2024, but remains stalled at present owing to an economic downturn in Finland's housing market. All projects from the first bidding have let go of their plot reservations. However, the tramline started functioning January 2025, bringing new people effortlessly to a formerly remote area.

The implementation phase of Hiedanranta integrated citizen-generated ideas – especially the Nordic superblock concept – into ambitious plot bidding criteria, demanding novel collaboration among designers and builders to realise shared spaces that support inclusion and civic resilience. Citizens' agency was further institutionalised through the creation of Hiedanrannan Palvelut Oy, a resident-owned service company enabling collective decision-making and community-driven use of shared spaces and digital platforms. Despite market challenges, the enduring identity of the area – shaped by citizen-led activities and collaborations – demonstrates how early civic engagement can influence long-term urban development frameworks. The analysis further demonstrates the concurrent necessity of diverse context-connecting practices to effectively foster civic resilience throughout different phases of ULLs.

6. CONTEXT-CONNECTING PRACTICES FOR CIVIC RESILIENCE

To understand how city authorities can enhance civic resilience (see Antaki & Petrescu 2023) during different phases of the district-sized ULL, the analysis focused on three context-connecting practices (Lehtimäki *et al.* 2023). The long-term interactions between city officials and citizens, beginning before even the project phase of the urban development (Artto *et al.* 2016), were examined. These practices varied across different phases and are summarised in Table 2, with the first two phases overlapping for a few years.

Table 2: The context-connecting practices city authorities used to promote civic resilience across different phases of Hiedanranta ULL.

CONTEXT-CONNECTING PRACTICE	PHASE 1: SETTING THE STAGE (2015–2019)	PHASE 2: COLLABORATIVE DESIGN (2017–2022)	PHASE 3: IMPLEMENTATION (2022–ONGOING)
Bringing actors and contexts together	Renting vacant spaces out cheaply with the demand of organising events that invite new people to the area and fostering practices that bring culture, innovation and planning together	Early co-design workshops and developing local collective spaces (*e.g.* public sauna) and inviting new kinds of citizens	Demanding, with legal contracts, that builders collaborate to create shared spaces for citizens on a block level and developing a digital platform for citizens (to share spaces, things, services; to support communality; to save energy)
Creating persistent tactics	Deliberate openness of the development process and recognising possible strong future actor groups to collaborate with (*e.g.* skateboarders)	Inviting citizens back several times to check on and further develop the plans and solidifying ideas to legal documents and creating citizen–expert coalitions	Establishing a new legal entity (Hiedanrannan Palvelut Oy) that owns and manages the shared spaces until new residents arrive and establishing efficient public transport early
Catalysing positive impact loops	Choosing actors that diversify social groups (*e.g.* Sopimusvuori) and brave adoption of new roles	Supporting emerged strong concepts (*e.g.* Nordic Superblock) that inspire varied actors	Residents will own and manage the new variety and quality shared spaces on a block level, deciding themselves how the use of spaces develop in future

In Hiedanranta, the practice of bringing actors and contexts together was foundational. In relation to citizens, during the first two phases, city authorities recognised citizens' expertise and trusted their ability to activate vacant spaces and co-design the area. City authorities believed that a shared vision would emerge as diverse actors collaborated on tasks they were most knowledgeable about. Time was allocated for the development of collaborations, learning and skill-building.

The city deliberately supported the mixing of actors through novel organisational structures, events and shared spaces. During the implementation phase, citizens were not directly involved but the city required builders to collaborate and form new legal structures. The city also reformed itself organisationally to realise the ideas that had emerged in the first two phases for the benefit of the citizens.

The practice of persistent tactics addresses the challenge noted in the system life-cycle model: how to retain people, practices and knowledge as the ULL transitions to a new phase. City authorities played a crucial role in identifying potential future collaborators early on, without prejudices, even without knowing the exact nature of the future collaboration. They also intentionally reopened planning processes that tend to shift from citizen involvement to professional oversight. While the goal was to solidify the best co-developed ideas into planning documents, this was not entirely successful owing to the differing scales of early planning documents and citizens' ideas. However, the city was actively questioning conventional practices behind the scenes, working to establish a needed new legal entity, Hiedanrannan Palvelut Oy. Managed by the city until new residents arrive, this entity ensures that shared spaces supporting community building will be available in the future phase when citizens move into the area.

The practice of catalysing positive impact loops extends over a longer timescale than the persistent tactics to support the stakeholder groups. The city's role was to identify weak signals, such as readiness for sharing spaces in a new way, and help them gain momentum. During the stage-setting phase, the mood and atmosphere of the area began to take shape. Inviting actors who empower vulnerable individuals into the area is an example of positive impact loops influencing the emerging identity of the area. The city also played a central role in embracing the concept of Nordic superblocks, co-developing it further, and reorganising its own practices to enable the realisation of this idea. It was crucial to ensure that future positive impact loops were facilitated, allowing future citizens to decide in detail what will occur in shared spaces and how they will collaborate.

In terms of civic resilience (Antaki & Petrescu 2023), the agency of citizens has grown in different phases. In the phase of setting the stage, many citizens learnt to take vacant spaces back into use in collaboration with the city. The activities that people brought into the spaces started to form the identity of the area. Citizens learnt that the future of the area is not yet decided and that there is a yearly event, the Hiedanranta garden party, in which the planning situation can always be discussed and the next opportunities to legally affect the plans are shared. The transparency of the yearly event had been initiated by the consulting company Kuowi Oy, and it had become a tradition as the city authorities felt they were getting a valuable reflection from it. In the phase of collaborative design, the citizens who participated to the workshops just after the results of the idea competition had been announced learnt to co-design with architects, planners and other designers. The agency of the citizens grew as some of the key concepts developed in the workshops gained momentum and became embedded to the design of the area. After the third phase, when people move to the area, they will have more agency in relation to the management and use of the shared spaces in their city blocks via the new legal entity that they will themselves also own.

7. DISCUSSION AND CONCLUSIONS

City authorities, such as urban planners and project workers, can increase civic resilience in the context of ULLs. To do so, they must first understand the dual context of change: the need of their organisation both to drive change and to change itself while doing so (Lehtimäki *et al.* 2023). This paper explored three context-connecting practices (Lehtimäki *et al.* 2023) that help them to do so. The novelty is on the focus of the context-connecting practices between citizens and city authorities, highlighting the potentials of this collaboration and the mediator role of the city to foster transition (Hernberg & Hyysalo 2024; Kivimaa *et al.* 2019).

Civic resilience is understood especially as citizens' learning capacity, with the focus on agency in the change processes. In the case study this became evident when the citizens learnt to take

vacant spaces back into use and when they learnt to work with architects and planners to initiate new design concepts. It is fundamental to understand that the growing agency of citizens was also needed from the city authorities' point of view: without capable and active people the spaces would not have been taken back into use, nor would the city and other designers alone have been able to push through the novel housing solutions. This highlights the importance of civic resilience for all actors in a network striving for change, opening a path to further examine questions of reciprocity, such as citizens using their free time to participate, while professionals get paid. It also points out how civic resilience is not an isolated island but an essential element in the broader concepts of resilience.

Results show that the practices the city authorities may successfully apply to support the increase of civic resilience vary greatly in different phases of a ULL. In early phases there are more background enabler roles that ask for an open mind and capacity to take risks. In later phases, in turn, the city authorities may be required to be the opposite: to be strong and stubborn to create new holding structures into their organisation that solidify the change already acquired. This role to stabilise new structures has also been noted as important in other transition studies (Kivimaa *et al.* 2019; Rohracher 2009). However, there is also the risk of stabilisation leading to institutionalisation, where the momentum for change is lost.

Further conceptual work is needed in future to understand the temporalities of civic resilience. Also, conceptual ULL models are needed that build on the temporalities that rise from the local legal, political and economic situations. The case illustrated how ULL phases were affected by these situations: the different temporalities of the planning process, the coalition to realise a new design idea and the economic cycle of the building sector (inhibiting implementation). A more developed temporal model would help city authorities with navigating the context-connecting.

The case study also revealed how important and powerful the collaboration with citizens is at the early stage before any exact plans exist. It demonstrates how the early collaborations contributed significantly to the formation of the area's identity and to the implementation of novel and radical design ideas.

While focusing on context-connecting practices such as bringing actors and contexts together, creating persistent tactics, and catalysing positive impact loops, the previously overlooked aspect of space becomes prominent. The catalytic power of taking vacant spaces back into use has been recognised in ULL literature (*e.g.* Puerari *et al.* 2018) but not as a context-connecting practice (Lehtimäki *et al.* 2023) and not in the system life-cycle view on building (Artto *et al.* 2016).

In Hiedanranta, the processes of taking vacant spaces back into use contributed significantly to the area development and its civic resilience. Vacant spaces can act as magnets, drawing different actors together and serving as incubators for new enterprises, cultural activities, and collaboration models between the city and citizens. When the city owns these premises, it can ensure favourable conditions for pioneers by keeping rents low and performing minimal renovations, while encouraging open events to foster interaction among early stakeholders. In co-design workshops, citizens expressed their need for shared spaces that support community building and civic resilience. These shared spaces are now being realised as the city facilitates new collaborations among builders, ensuring that spaces supporting civic resilience are integrated into the future built environment.

Context-connecting practices (Lehtimäki *et al.* 2023) and a system life-cycle view (Artto *et al.* 2016) add to community-based understandings of ULLs (*e.g.* Petrescu *et al.* 2016) by proving that citizens are crucial stakeholders (see also Moulaert *et al.* 2010) in complex processes such as realising progressive city districts, from an early stage. Citizens are not only involved but are actually needed by other stakeholders to ensure the success of the project. This accentuates the presence of civic resilience in the core of the general idea of resilience (Othengrafen *et al.* 2024: 24–25).

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All authors have contributed to the data collection, analyses and writing. MK: introduction and case descriptions. EA: analytical lenses, discussion and conclusions. ML: supervision and funding.

COMPETING INTERESTS

The authors have no competing interests to declare.

DATA AVAILABILITY

Interview data are not publicly available.

ETHICAL APPROVAL

Ethics committee approval was not required. All interviewees and participants in recorded discussions signed an informed consent.

FUNDING

This research was funded by Research Council of Finland projects 'Contested Waterfront Transformations' (grant number 363767) and 'Visions of the City' (grant number 351019), and by the Business Finland project 'The Nordic Superblocks as Decarbonization Catalysts' (grant number 3038/31/2023).

REFERENCES

Alatalo, E., Kuoppa, J., Kyrönviita, M., Laine, M., & **Karppi, L.** (2017). *From ideas competition to citizen's vision. Planning Hiedanranta in follow-on workshops*. Tampere: The City of Tampere. https://www.e-julkaisu.fi/tampereen_kaupunki/from-ideas-competition-to-citizens-visions/mobile.html#pid=1

Alatalo, E., Kyrönviita, M., & **Leino, H.** (2019). *Tyhjät tilat kaupunkien voimavaraksi!* Politiikkasuositus. [Vacant spaces as resources of the city! Policy brief]. Tampereen yliopisto, Tampere. https://www.aka.fi/globalassets/3-stn/1-strateginen-tutkimus/tiedon-kayttajalle/politiikkasuositukset/politiikkasuositukset/19_2_tyhjat_tilat_kaupunkien_voimavaraksi.pdf

Alatalo, E., Leino, H., Laine, M., & **Turku, V.** (2023a). Heating up the sauna: Analogue model unraveling the creativity of public participation. *Planning Theory*, 23(1), 22–41. https://doi.org/10.1177/14730952231166567

Alatalo, E., Turku, V., Milián Bernal, D., & **Kyrönviita, M.** (2023b). Learning urban activism: A retrospective look at how to engage students. *DATUTOP*, 41, 169–194. https://urn.fi/URN:ISBN:978-952-03-3007-1

Antaki, N., & **Petrescu, D.** (2023). Designers' roles in civic pedagogies of co-making: Lessons from the global south and north. *CoDesign*, 19(1), 51–73. https://doi.org/10.1080/15710882.2022.2123927

Artto, K., Ahola, T., & **Vartiainen, V.** (2016). From the front end of projects to the back end of operations: Managing projects for value creation throughout the system lifecycle. *International Journal of Project Management*, 34, 258–270. https://doi.org/10.1016/j.ijproman.2015.05.003

Baxter, J.-S. (2022). From urban living labs towards critical spatial design: Decolonising knowledge in urban design and planning using critical reflexivity. *Nordic Journal of Urban Studies*, 2(2), 124–140. https://doi.org/10.18261/njus.2.2.3

Braun, V., & **Clarke, V.** (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Bulkeley, H., & **Betsill, M. M.** (2013). Revisiting the urban politics of climate change. *Environmental Politics*, 22(1), 136–154. https://doi.org/10.1080/09644016.2013.755797

Bulkeley, H., Marvin, S., Palgan, Y. V., McCormick, K., Breitfuss-Loidl, M., Mai, L., von Wirth, T., & **Frantzeskaki, N.** (2019). Urban living laboratories: Conducting the experimental city? *European Urban and Regional Studies*, 26(4), 317–335. https://doi.org/10.1177/0969776418787222

Carmona, M. (2009). Design coding and the creative, market and regulatory tyrannies of practice. *Urban Studies*, 46(12), 2643–2667. https://doi.org/10.1177/0042098009344226

Chevalier, J. M. (2019). *Participatory action research: Theory and methods for engaged inquiry* (2nd ed.). Routledge. https://doi.org/10.4324/9781351033268

Clark, A., Holland, C., Katz, J., & **Peace, S.** (2009). Learning to see: Lessons from a participatory observation research project in public spaces. *International Journal of Social Research Methodology*, 12(4), 345–360. https://doi.org/10.1080/13645570802268587

Crossan, J., Cumbers, A., McMaster, R., & **Shaw, D.** (2016). Contesting neoliberal urbanism in Glasgow's community gardens: The practice of DIY citizenship. *Antipode*, 48(4), 937–955. https://doi.org/10.1111/anti.12220

De Lille, C. S. H., & **Overdiek, A.** (2021). From system to local to system: Design principles to scale for a system in transition. *Proceedings of Relating Systems Thinking and Design, RSD10*. https://rsdsymposium.org/system-design-principles-to-scale/

Fabian, L., & **Samson, K.** (2015). Claiming participation – a comparative analysis of DIY urbanism in Denmark. *Journal of Urbanism: International Research on Placemaking and Urban Sustainability*, 9(2), 166–184. https://doi.org/10.1080/17549175.2015.1056207

Frantzeskaki, N., Broto, V. C., Coenen, L., & **Loorbach, D.** (2017). *Urban sustainability transitions*. Routledge. https://doi.org/10.4324/9781315228389

Hemphill, M. A., & **Richards, K. A. R.** (2018). A practical guide to collaborative qualitative data analysis. *Journal of Teaching in Physical Education*, 37(2), 225–231. https://doi.org/10.1123/jtpe.2017-0084

Hernberg, H., & **Hyysalo, S.** (2024). Modes of intermediation: How intermediaries engage in advancing local bottom-up experimentation. *Environmental Innovation and Societal Transitions*, 51, 100849. https://doi.org/10.1016/j.eist.2024.100849

Juujärvi, S., & **Pesso, K.** (2013). Actor roles in an urban living lab: What can we learn from Suurpelto, Finland? *Technology Innovation Management Review*, 3(11): 22–27. http://doi.org/10.22215/timreview/742

Kivimaa, P., Boon, W., Hyysalo, S., & **Klerkx, L.** (2019). Towards a typology of intermediaries in sustainability transitions: A systematic review and a research agenda. *Research Policy*, 48(4), 1062–1075. https://doi.org/10.1016/j. respol.2018.10.006

Korpinen, V., & **Pulkkinen, R.** (2015). *Hiedanrannan ihmiset*. [*The people of Hiedanranta*] Kuowi Oy & Tampereen kaupunki. https://www.slideshare.net/slideshow/hiedanrannan-ihmiset/58434043

Kronsell, A., & **Mukhtar-Landgren, D.** (2018). Experimental governance: The role of municipalities in urban living labs. *European Planning Studies*, 26(5), 988–1007. https://doi.org/10.1080/09654313.2018.1435631

Kyrönviita, M., & **Leino, H.** (2025). Fulfill the dream: Advancing skateboarders' agency through experimenting and interaction with urban infrastructure. *Journal of Urbanism*, 1–20. https://doi.org/10.1080/17549175.2025.2507320

Kyrönviita, M., & **Wallin, A.** (2022). Building a DIY skatepark and doing politics hands-on. *City*, 26(4), 646–663. https://doi.org/10.1080/13604813.2022.2079879

Lehtimäki, H., Jokinen, A., & **Pitkänen, J.** (2023). Erratum to 'Project-based practices for promoting a sustainability transition in a city organization and its urban context' [Int. J. Proj. Manag., 41 (2023), 102516]. *International Journal of Project Management*, 41(8), 102539. https://doi.org/10.1016/j.ijproman.2023.102539

Lehtovuori, P., Edelman, H., Rintala, J., Jokinen, A., Rantanen, A., Särkilahti, M., & **Joensuu, T.** (2016). *Hiedanrannan kehittämisvisio: Tiivis ja intensiivisesti vihreä Tampere City West*. [Hiedanranta vision: the compact and intensively green Tampere City West] Tampereen teknillinen yliopisto, Arkkitehtuurin laitos. https://trepo.tuni.fi//handle/10024/116771

MacKinnon, D., & **Derickson, K. D.** (2013). From resilience to resourcefulness: A critique of resilience policy and activism. *Progress in Human Geography*, 37(2), 253–270. https://doi.org/10.1177/0309132512454775

Madanipour, A. (2018). Temporary use of space: Urban processes between flexibility, opportunity and precarity. *Urban Studies*, 55(5), 1093–1110. https://doi.org/10.1177/0042098017705546

Magis, K. (2010). Community resilience: An indicator of social sustainability. *Society & Natural Resources*, 23(5), 401–416. https://doi.org/10.1080/08941920903305674

Morris, P. W. G. (2013). *Reconstructing project management*. John Wiley & Sons. https://doi.org/10.1002/9781118536698

Moulaert, F., Swyngedouw, E., Martinelli, F., Gonzalez, S. (eds.) (2010). *Can neighbourhoods save the city? Community development and social innovation*. Routledge. https://doi.org/10.1080/15575330.2025.2498835

Mukhtar-Landgren, D., Kronsell, A., Voytenko Palgan, Y., & **von Wirth, T.** (2019). Municipalities as enablers in urban experimentation. *Journal of Environmental Policy & Planning*, 21(6), 718–733. https://doi.org/10.1080/1523908X.2019.1672525

Okely, J. (2012). *Anthropological practice: Fieldwork and the ethnographic method*. Routledge. https://doi.org/10.4324/9781003084563

Othengrafen, F., Ziehl, M., Herrmann, S., & **Pencic, D.** (2024). Community resilience: Transformative capacity as driver for social cohesion and sustainable development. In F. Othengrafen, S. Herrmann, D. Pencic and S. Lazarevski (Eds.), *Social cohesion and resilience through citizen engagement* (pp. 19–33). Edward Elgar. https://doi.org/10.4337/9781035317103.00011

Overdiek. (2024). Citylab X: Towards a lens on the urban living lab as driver of systemic innovation. Contexts. *The Systemic Design Journal*, 2. https://doi.org/10.58279/v2002

Petrescu, D., Petcou C., & **Baibarac, C.** (2016). Co-producing commons-based resilience: Lessons from R-Urban. *Building Research & Information*, 44(7), 717–736. https://doi.org/10.1080/09613218.2016.1214891

Portelli, J. P., & **McMahon, B. J.** (2004). "Why critical-democratic engagement?" *Journal of Maltese Education Research*, 2(2), 39–45. https://www.um.edu.mt/library/oar/bitstream/123456789/18926/1/2.%20WHY%20CRITICAL-DEMOCRATIC%20ENGAGEMENT.pdf

Puerari, E., De Koning, J. I., Von Wirth, T., Karré, P. M., Mulder, I. J., & **Loorbach, D. A.** (2018). Co-creation dynamics in urban living labs. *Sustainability*, 10(6), 1893. https://doi.org/10.3390/su10061893

Rikala, S., Wallin, A., & **Sjöblom, J.** (2023). Miten sosiaalista kestävyyttä tehdään? Ruohonjuuritason toimijoiden sosiaaliset käytännöt entisen tehdasalueen elävöittämisessä. [How social sustainability is made? Social practices of grassroots organizations in activating a formal factory area] *Sosiologia*, 60(1), 5–21. https://journal.fi/sosiologia/article/view/128681

Rizzo, A., Ekelund, B., Bergström, J., & **Ek, K.** (2020). Participatory design as a tool to create resourceful communities in Sweden. In C. Smaniotto Costa, M. Mačiulienė, M. Menezes & B. Goličnik Marušić (Eds.), *Co-creation of public open places. Practice – reflection – learning* (pp. 95–108). Lusófona University Press. https://doi.org/10.24140/2020-sct-vol.4

Robazza, G. (2024). Public life on stage: Urban living labs as cultural performative spaces. In A. Melis, R. Vavetsi & A. Chisty (Eds.), *The architecture of exhibitions: Experiential design*. Routledge.

Rohracher, H. (2009). Intermediaries and the governance of choice: The case of green electricity labelling. *Environmental Planning A*, 41, 2014–2028. https://doi.org/10.1068/a41234

Sjöblom, J., Kuoppa, J., Laine, M., & **Alatalo, E.** (2021). Crafting a planning issue with citizens in the context of planning competition: a case of 'Nordic Superblock'. *Journal of Urban Design*, 26(1), 117–131. https://doi.org/10.1080/13574809.2020.1832886

Statistics Finland. (2025). Statistics Finland's free-of-charge statistical databases. Preliminary population structure by area, 2024M01*-2024M01*. https://pxdata.stat.fi/PxWeb/pxweb/en/StatFin/StatFin__vamuu/statfin_vamuu_pxt_11lj.px

The City of Tampere. (2016). Competition programme, The Hiedanranta district of Tampere – international ideas competition. https://www.safa.fi/wp-content/uploads/2016/09/2016_Tampere_hiedanranta_kilpailuohjelma_competitionprogramme.pdf

The City of Tampere. (2017a). Evaluation minutes, The Hiedanranta district of Tampere – international ideas competition. https://www.safa.fi/wp-content/uploads/2016/09/2016_Tampereen_Hiedanranta_Arvostelupoytakirja_eng_fi.pdf

The City of Tampere. (2017b). Hiedanranta Structure Plan. https://www.tampere.fi/sites/default/files/2022-05/hiedanranta_rakennesuunnitelma.pdf

The City of Tampere. (2020). Hiedanranta Master Plan. https://www.tampere.fi/sites/default/files/2022-05/hiedanrannan-yleissuunnitelma-saavutettava-raportti-englanniksi.pdf

The City of Tampere. (2025). Hiedanrannan alueen suunnittelu ja kaavoitus. [The design and planning of Hiedanranta] https://www.tampere.fi/kaupunkisuunnittelu/kaupunkiymparisto-uudistuu/hiedanranta/hiedanrannan-alueen-suunnittelu-ja-kaavoitus

Turku, V., Jokinen, A., & **Jokinen, P.** (2022). How do time-bound practices initiate local sustainability pathways? *Sustainable Cities and Society*, 79, 103697–. https://doi.org/10.1016/j.scs.2022.103697

Virkkala, I. (2016). *Hiedanrannan talkoot*. [Making Hiedanranta together] Uusi kaupunki kollektiivi, Kuowi Oy & Tampereen kaupunki.

TO CITE THIS ARTICLE:
Alatalo, E., Laine, M., & Kyrönviita, M. (2025). Increasing civic resilience in urban living labs: city authorities' roles. *Buildings and Cities*, 6(1), pp. 917–933. DOI: https://doi.org/10.5334/bc.610

Submitted: 14 April 2025
Accepted: 29 October 2025
Published: 18 November 2025

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